CLINICAL · EAR

Perichondritis, zoster and pinna infections

A red, hot, swollen pinna that spares the lobule is a disease of the cartilage, because the lobule has none. Perichondritis needs anti-pseudomonal cover and early drainage of any pus; otherwise the cartilage dies and the ear crumples. Vesicles in the concha with facial weakness are a different emergency: Ramsay Hunt syndrome.

AT A GLANCE

- Red, hot pinna that spares the lobule means cartilage.
- Cover Pseudomonas and drain early.
- Zoster: vesicles, pain, and often facial palsy.
- Steroids plus antivirals for Ramsay Hunt.

1 CHAPTER ONE Overview

Infections and inflammation of the pinna.

1.1 The conditions

RED, PAINFUL PINNA

- SKIN **Cellulitis, erysipelas** staphylococci and streptococci
- CARTILAGE **Perichondritis and chondritis** Pseudomonas
- VIRUS **Herpes zoster oticus** Ramsay Hunt syndrome
- AUTOIMMUNE **Relapsing polychondritis** also the nose, larynx and trachea
- DEGENERATIVE **Chondrodermatitis nodularis helicis** painful nodule on the helix

1.2 Applied anatomy

The cartilage of the pinna has no blood supply of its own and lives on its perichondrium. When pus or blood separates the two, the cartilage starves and necroses. The lobule has no cartilage, so a cartilage disease spares it.

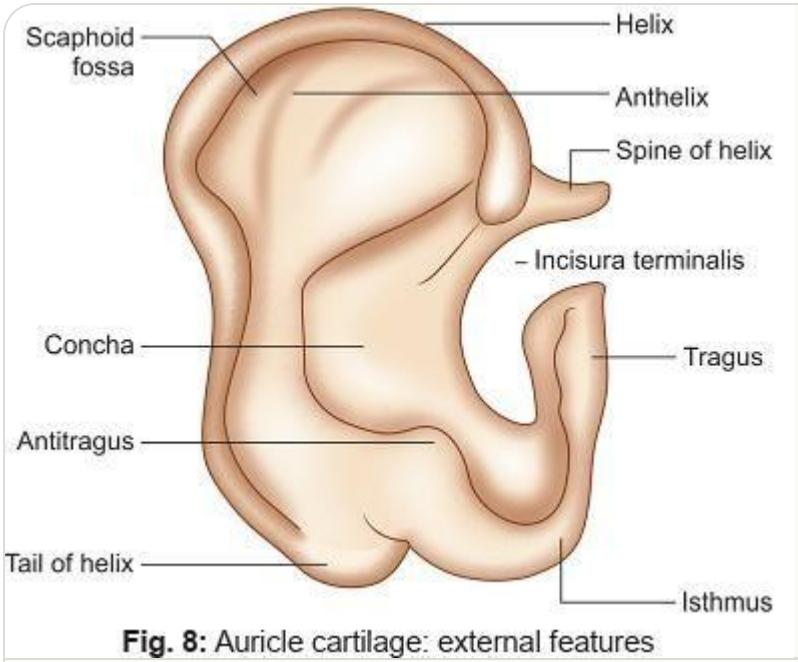


FIGURE 1 The auricular cartilage: everything except the lobule.
Anatomy atlas · ENT Steps p. 29

CHAPTER TWO

2 Perichondritis

Cartilage under threat.

2.1 Causes

Blunt trauma with an infected haematoma, lacerations and burns, high (cartilage) piercing, ear surgery, and spread from otitis externa. *Pseudomonas aeruginosa* is the usual organism, with staphylococci.

2.2 Clinical picture

2.2.1 Early

Pain and a red, hot, tender, thickened pinna, sparing the lobule.

2.2.2 Late

Fluctuant swelling as pus collects between perichondrium and cartilage, then necrosis.



FIGURE 2 Perichondritis of the pinna: the whole pinna is red and swollen.
Dhingra, fig. 8.10

2.3 Treatment

2.3.1 Early, without abscess

An oral fluoroquinolone such as ciprofloxacin, or intravenous anti-pseudomonal antibiotics for severe infection; cool aluminium acetate compresses.

2.3.2 Abscess

Incise along a natural fold, remove dead cartilage, irrigate, and apply a bolster dressing for several days to keep the skin against the cartilage. Antibiotics continue for two to four weeks.

2.4 Outcome

Loss of cartilage leaves a shrivelled, deformed *cauliflower ear*.



FIGURE 3 A cauliflower ear, the end result of cartilage necrosis after haematoma or perichondritis.

Dhingra, fig. 8.7

3 CHAPTER THREE Cellulitis and erysipelas

Skin infections.

3.1 Cellulitis

Spreading skin infection, usually staphylococcal or streptococcal, often after a scratch, piercing or otitis externa. Unlike perichondritis, it may involve the lobule.

3.2 Erysipelas

Streptococcal infection with a sharply defined, raised red edge, which can spread onto the face. Treated with penicillin.

4 CHAPTER FOUR Herpes zoster oticus

Ramsay Hunt syndrome.

4.1 Cause

Reactivation of varicella-zoster virus in the geniculate ganglion.

4.2 Clinical picture

Severe, burning ear pain, followed by vesicles in the concha, canal and sometimes the palate and tongue, then facial palsy. The eighth nerve is often involved, with tinnitus, sensorineural hearing loss and vertigo.



FIGURE 4 Ramsay Hunt syndrome: facial palsy with small vesicles in the right concha.
Dhingra, fig. 14.9

4.3 Treatment

High-dose prednisolone (about 1 mg/kg for 14 days) with an antiviral such as valaciclovir, analgesia, and eye protection. Gabapentin for post-herpetic neuralgia.

Surgical decompression is not advised, because the nerve damage is diffuse.

4.4 Outlook

Worse than Bell's palsy: only about 60% regain normal facial function.

OPERATIVE GUIDES IN THEATRE 2 operations

Eyelid Reanimation in Patients With Facial Paralysis
Eyelid reanimation

Facial Reanimation
Facial reanimation

5 CHAPTER FIVE Inflammatory conditions

Not infections.

5.1 Relapsing polychondritis

An autoimmune disease of cartilage: recurrent painful red swelling of the pinna sparing the lobule, often both ears, with nasal (saddle nose), laryngeal and tracheal cartilage involvement that can threaten the airway. Treated with high-dose steroids and immunosuppression.

5.2 Chondrodermatitis nodularis helicis

A painful, tender nodule on the rim of the helix, usually in older men who sleep on that side. Relieve pressure; excise if persistent, and send for histology to exclude skin cancer.

5.3 Differential diagnosis

Gout (tophi on the helix), lymphoma and skin tumours.

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 On call

A RED, SWOLLEN EAR

1. Does it spare the lobule? Then it is cartilage.
2. Fluctuant: drain under anaesthesia and bolster.
3. Start anti-pseudomonal antibiotics.
4. Vesicles with facial weakness: steroids, antivirals and eye protection.
5. Eye symptoms: check the cornea.

6.2 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Why the lobule is spared.**
- **Pseudomonas** as the usual organism.
- **Drainage and bolster,** and why cartilage dies.
- **Ramsay Hunt:** vesicles, palsy, eighth nerve involvement, treatment and prognosis.
- **Relapsing polychondritis** and the airway.

6.3 Common pitfalls

COMMON PITFALLS

- Using flucloxacillin alone for perichondritis.
- Leaving pus under the perichondrium.

- Calling Ramsay Hunt syndrome Bell's palsy because nobody looked in the ear.

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